DISCRETE STRUCTURES

CMSC250

Chapter 10: Constructive Proofs and Proof by Cases

By Naren Ayinala

Existence, witnesses, exhaustion, and case analysis



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1 Constructive proofs: the witness is the proof

An existential claim $\exists x \in D\, P(x)$ can be proved by giving a particular $x \in D$ and checking $P(x)$. Constructive proofs often begin "Let $n = \dots$" and finish "Thus the chosen value satisfies the claim. ■" A witness must meet every condition, including domain and distinctness restrictions.

Key idea. For an existential statement, you do not need to explain how you discovered the witness. You must verify that it works.

2 Worked constructive proofs

Worked example: three distinct natural numbers

Claim. There exist distinct $a, b, c \in \mathbb{N}$ with $a + b^2 = c^3$.

Proof. Let $a = 0$, $b = 8$, and $c = 4$. These are natural numbers under the course convention that $0 \in \mathbb{N}$, and they are pairwise distinct. Moreover,

$$a + b^2 = 0 + 8^2 = 64 = 4^3 = c^3.$$

Thus this triple is a witness for the existential claim. ■

Why this works. It is essential to check *both* the equation and the requirement that all three values are distinct.

Worked example: a composite number of a specified form

Claim. There exists $n \in \mathbb{N}$ such that $3^n + 2$ is composite.

Proof. Choose $n = 5$. Then

$$3^5 + 2 = 243 + 2 = 245 = 5 \cdot 49.$$

Both 5 and 49 are nontrivial positive factors, so 245 is composite. Since $n = 5 \in \mathbb{N}$, the claim holds. ■

Worked example: dividing a product without dividing either factor

Claim. There exist natural numbers a, b, c such that $a \mid bc$, but $a \nmid b$ and $a \nmid c$.

Proof. Choose $a = 6$, $b = 2$, and $c = 3$. Since $bc = 6$, we have $6 \mid bc$. But $6 \nmid 2$ and $6 \nmid 3$, as neither 2 nor 3 is an integer multiple of 6. All three are natural numbers. Thus the witness satisfies every condition. ■

Worked example: two different prime decompositions

Claim. There is a natural number expressible as a sum of three distinct primes in two different ways.

Proof. Choose $n = 23$. We have

$$23 = 3 + 7 + 13 = 5 + 7 + 11.$$

In each sum, all three summands are distinct primes, and the two collections differ. Hence 23 is a witness. ■

Worked example: a square plus 49 is a cube

Claim. There exist natural numbers a, b such that $a^2 + 49 = b^3$.

Proof. Choose $a = 524$ and $b = 65$. Then

$$a^2 + 49 = 274576 + 49 = 274625 = 65^3 = b^3.$$

Both values are natural numbers, so this is a valid witness. ■

Why this works. For equations of this kind, finding a witness may require exploration, but the written proof only needs the correct values and verification.

3 How to find a witness efficiently

Start from the constraints, not random guesses. For $a^2 + 49 = b^3$, the equation says $a^2 = b^3 - 49$; test cubes greater than 49 and ask whether their difference is a square. For divisibility witnesses, small composite values such as $6 = 2 \cdot 3$ often help. For claims requiring distinct primes, list a few small primes and test sums.

4 Universal generalization and constructive witnesses

A claim of the form $\forall n \in D\, \exists m \in E\, P(n, m)$ requires a *rule* producing a witness m for each arbitrary n. The witness may depend on n.

Worked example: every integer has a larger integer

Claim. For every $n \in \mathbb{Z}$, there is an $m \in \mathbb{Z}$ with $m > n$.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. Choose $m = n + 1$. By closure of $\mathbb{Z}$ under addition, $m \in \mathbb{Z}$, and clearly $m > n$. Since n was arbitrary, the claim holds. ■

Worked example: every integer has a larger even integer

Claim. For each integer n, there is an even integer m with $m > n$.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. Choose $m = 2(|n| + 1)$. This is even because $|n| + 1 \in \mathbb{Z}$. Also $m > |n| \geq n$, so $m > n$. Thus the required even integer exists for every n. ■

5 Proof by exhaustion: check every element

When a universal statement has a small *finite* domain, a table of every input can prove it. State why your list is complete; checking examples from an infinite domain is not exhaustion.

Worked example: a prime-valued expression on a finite domain

Claim. For integers n with $1 \le n \le 10$, the number $n^2 - n + 11$ is prime.

Proof. The allowed integers are exactly $1, 2, \ldots, 10$. Their values are

n	1	2	3	4	5	6	7	8	9	10
$n^2 - n + 11$	11	13	17	23	31	41	53	67	83	101

Each listed value is prime (divisibility can be checked by primes not exceeding its square root). These ten cases exhaust the domain, so the claim follows. ■

Why this works. Showing only $n = 1, 2, 3$ would be evidence, not a proof. Exhaustion requires all ten values.

Worked example: no primes strictly between 113 and 127

Claim. There is no prime integer strictly between 113 and 127.

Proof. The integers strictly between these endpoints are $114, 115, \ldots, 126$. Each is composite:

n	114	115	116	117	118	119	120
factor	2	5	2	3	2	7	2

n	121	122	123	124	125	126
factor	11	2	3	2	5	2

Each displayed factor is greater than 1 and smaller than its number. Since these are all integers in the interval, none is prime. ■

6 Proof by cases: a finite logical partition

Proof by cases differs from exhaustion: each case may contain infinitely many objects. The key is to cover the domain by logical categories, such as even/odd or positive/zero/negative.

Worked example: a polynomial is always odd

Claim. For every integer n, $n^2 + n + 3$ is odd.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. We consider two cases.

Case 1: n is even. Then $n = 2k$ for $k \in \mathbb{Z}$, and

$$n^2 + n + 3 = 4k^2 + 2k + 3 = 2(2k^2 + k + 1) + 1.$$

The parenthesized quantity is an integer by closure, so the expression is odd.

Case 2: n is odd. Then $n = 2k + 1$ for $k \in \mathbb{Z}$, and

$$\begin{aligned} n^2 + n + 3 &= (2k+1)^2 + (2k+1) + 3 \\ &= 4k^2 + 6k + 5 \\ &= 2(2k^2 + 3k + 2) + 1. \end{aligned}$$

Again the parenthesized quantity is an integer, so the expression is odd.

Every integer is either even or odd, so the conclusion holds for all integers. ■

Worked example: the product of consecutive integers is even

Claim. For all integers n, $n(n+1)$ is even.

Proof. Let $n \in \mathbb{Z}$ be arbitrary. If n is even, write $n = 2k$, so $n(n+1) = 2[k(n+1)]$ and closure gives an integer witness. If n is odd, write $n = 2k+1$, so $n+1 = 2(k+1)$ and $n(n+1) = 2[n(k+1)]$, again with an integer witness. These cases cover every integer, so $n(n+1)$ is even for all integers n. ■

Worked example: absolute values and sign cases

Claim. For all real numbers x, $|x| \geq x$.

Proof. Let $x \in \mathbb{R}$ be arbitrary. If $x \geq 0$, then $|x| = x$, so the inequality holds with equality. If $x < 0$, then $|x| = -x > x$. These cases are exhaustive, so $|x| \geq x$ for every real x. ■

7 When to use which method

Claim shape	Best starting point
$\exists x\, P(x)$	Find and verify one witness.
$\forall x \in \{a, b, c\}\, P(x)$	Check all three inputs; state completeness.
$\forall n \in \mathbb{Z}\, P(n)$ with parity	Split into even and odd.
$\forall x \in \mathbb{R}\, P(\lvert x \rvert)$	Split into $x \geq 0$ and $x < 0$.
$\forall n\, \exists m\, P(n, m)$	Choose arbitrary n, construct m in terms of n.

8 Frequent errors and grading checklist

1. Did you verify the witness belongs to the domain (natural, integer, positive, prime)?
2. If values must be distinct, did you check distinctness?
3. If claiming composite, did you exhibit nontrivial factors?
4. If proving a universal statement by exhaustion, did you check every input?
5. If using cases, did you say why they cover every possible input?
6. If writing $2k$ or $2k+1$, did you say k is an integer and cite closure where needed?
7. Did you write a conclusion before the tombstone ■?

9 Practice problems

1. Construct $a, b, c \in \mathbb{N}$ with $a \mid bc$ but $a \nmid b$ and $a \nmid c$.
2. Find an integer n such that $n^2 - 1$ is a positive multiple of 8.
3. Prove $\forall n \in \mathbb{Z}\, \exists m \in \mathbb{Z}\, (m < n)$ constructively.
4. Prove by cases that $n^2 + n$ is even for all integers n.
5. Prove by cases that $|x| \geq 0$ for all real x.
6. Prove by exhaustion that $n^2 + n + 2$ is even for $n \in \{0, 1, 2, 3, 4\}$.
7. Find a counterexample to "every positive integer has a smaller positive integer."
8. Explain why testing ten positive integers does not prove a claim about *all* positive integers.

10 Detailed practice solutions

1. Choose $a = 6$, $b = 2$, $c = 3$. Then $6 \mid 6$ but $6 \nmid 2$ and $6 \nmid 3$. All values are natural numbers. ■
2. Choose $n = 3$. Then $n^2 - 1 = 8 = 8 \cdot 1$, a positive multiple of 8. ■
3. Let $n \in \mathbb{Z}$ be arbitrary. Choose $m = n - 1$. By integer closure, $m \in \mathbb{Z}$, and $m < n$. The claim follows. ■
4. If $n = 2k$, then $n^2 + n = 2(2k^2 + k)$, even by closure. If $n = 2k + 1$, then $n^2 + n = 2(2k^2 + 3k + 1)$, also even. Every integer is even or odd, so the claim holds. ■
5. If $x \geq 0$, then $|x| = x \geq 0$. If $x < 0$, then $|x| = -x > 0$. These cases exhaust the real numbers. ■
6. For $n = 0, 1, 2, 3, 4$, the values $n^2 + n + 2$ are respectively $2, 4, 8, 14, 22$, all even. These are all the elements of the specified domain. ■
7. The positive integer 1 has no smaller positive integer. This disproves the universal claim. ■
8. The set of positive integers is infinite. Ten examples leave infinitely many untested inputs, so the universal conclusion does not follow.